Covariance summarizes joint variation

For a vector $\boldsymbol{x}$ with mean $\mu = E[\boldsymbol{x}]$

$$\Sigma_{ij} = \operatorname{Cov}[X_i, X_j]$$
$$= E[(X_i - \mu_i)(X_j - \mu_j)]$$

$$\Sigma = E[(\boldsymbol{x} - \mu)(\boldsymbol{x} - \mu)^T]$$
$$= E[\boldsymbol{x}\boldsymbol{x}^T] - \mu\mu^T$$

For three coordinates

$$\Sigma = \begin{bmatrix} V[X_1] & \operatorname{Cov}[X_1, X_2] & \operatorname{Cov}[X_1, X_3] \\ \operatorname{Cov}[X_2, X_1] & V[X_2] & \operatorname{Cov}[X_2, X_3] \\ \operatorname{Cov}[X_3, X_1] & \operatorname{Cov}[X_3, X_2] & V[X_3] \end{bmatrix}$$

Murphy, Book 1 §3.1.1, Eqs. 3.1–3.4

- Positive / negative covariance: deviations tend to have the same / opposite signs
- Diagonal entries are variances; off-diagonal entries are covariances
- Zero covariance does not imply independence

Conditioning shifts the mean by an observed deviation

$$p(x_1 \mid x_2) = \mathcal{N}\left(x_1 \mid \mu_{1|2}, \Sigma_{1|2}\right)$$

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$

Mean proof

$$B = \Sigma_{12}\Sigma_{22}^{-1}$$

$$r = x_1 - \mu_{1|2} = x_1 - \mu_1 - B(x_2 - \mu_2)$$

$$E[r] = 0,$$

$$\mathrm{Cov}[r, x_2] = \Sigma_{12} - B\Sigma_{22}$$

$$= 0$$

(r, x_2) is jointly Gaussian:

zero cross-covariance gives independence

$$E[r \mid x_2] = E[r] = 0$$

$$E[x_1 \mid x_2] = \mu_{1|2}$$

Murphy, Book 2 §2.3.1.3, Eq. 2.78; proof uses affine Gaussian closure

- Conditioning shifts the mean according to the observed deviation $x_2 - \mu_2$, cross-covariance, and observed variance

Bayes' rule for Gaussians

Model and joint Gaussian

$$\begin{pmatrix} z \\ y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_z \\ \mu_y \end{pmatrix}, \begin{pmatrix} \Sigma_{zz} & \Sigma_{zy} \\ \Sigma_{yz} & \Sigma_{yy} \end{pmatrix} \right)$$

$$\mu_y = W\mu_z + b, \quad \Sigma_{zz} = \Sigma_z$$
$$\Sigma_{zy} = \Sigma_z W^T = \Sigma_{yz}^T$$
$$\Sigma_{yy} = W\Sigma_z W^T + \Omega$$

Gaussian conditioning rule

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$
$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

Apply: $x_1 = z, x_2 = y$

$$p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z|y}, \Sigma_{z|y}\right)$$

$$\mu_{z|y} = \mu_z + \Sigma_{zy}\Sigma_{yy}^{-1}\left(y - \mu_y\right)$$

$$\Sigma_{z|y} = \Sigma_{zz} - \Sigma_{zy}\Sigma_{yy}^{-1}\Sigma_{yz}$$

Murphy, Book 2 §2.3.2.2, Eq. 2.121

- Apply the Gaussian conditioning rule to the joint blocks
- The mean responds to observed minus predicted measurement
- The covariance subtracts the uncertainty explained by the measurement

The scalar Gaussian update

Carried forward Earlier: Gaussian update formula

$$\mu_{z|y} = \mu_z + K(y - \mu_y)$$
$$K = \Sigma_{zy}\Sigma_{yy}^{-1}$$

Scalar specialization

$$y, z \in \mathbb{R}, \quad W = 1, \quad b = 0$$
$$\Sigma_z = \sigma_z^2, \quad \Omega = \sigma_\varepsilon^2$$
$$z \sim \mathcal{N}(\mu_z, \sigma_z^2)$$
$$y = z + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$$

ε is independent of z

Predicted mean

$$\mu_y = E[z + \varepsilon] = \mu_z$$

Cross-covariance

$$\Sigma_{zy} = \mathrm{Cov}[z, z + \varepsilon] = \sigma_z^2$$

Measurement variance

$$\Sigma_{yy} = \mathrm{Var}[z + \varepsilon] = \sigma_z^2 + \sigma_\varepsilon^2$$

$$K = \frac{\sigma_z^2}{\sigma_z^2 + \sigma_\varepsilon^2} \quad \rightarrow \quad \mu_{z|y} = \mu_z + K(y - \mu_z)$$

Murphy, Book 2 §2.3.2.2; scalar specialization of Eqs. 2.121–2.123

- The scalar gain is prior state variance divided by total measurement variance